Label Budget and Sensor-Outage Sensitivity of Masked Reconstruction in Human Activity Recognition

14 October 2026 | Empirical study for advisor review

Abstract

We compare supervised training with three masked-reconstruction pretraining objectives on subject-disjoint UCI HAR data. Forty-eight training configurations cover four label budgets and three random seeds; each checkpoint is evaluated under three signal conditions. At a 1% training-label budget, supervised training obtains 68.52 +/- 0.84% Macro-F1, temporal masking obtains 72.00 +/- 1.47% (+3.48 percentage points), and combined temporal-channel masking obtains 68.95 +/- 2.06%. These findings apply to one fixed subject split and do not establish a new algorithm or state-of-the-art performance. Full validation labels are used for selection and are excluded from the stated training-label budgets.

1. Research question

Wearable human activity recognition maps multichannel inertial measurements to activity labels. This study asks whether a compact encoder pretrained to reconstruct masked measurements improves classification when training labels are limited, and whether any improvement persists after a complete sensor group is suppressed. The contribution is a transparent experimental comparison with paired label subsets and explicit annotation-budget accounting.

2. Positioning against prior work

TS-TCC learns time-series representations through temporal and contextual contrastive tasks [2]. TS2Vec uses hierarchical contrasting [3]. Channel and combined time-channel masking have already been investigated for HAR [4], and RobustHAR studies multiscale masked pretraining for missing-data robustness [5]. Consequently, neither masking across time and channels nor studying sensor outages is claimed as a new idea here. Our independently written reconstruction baselines are not reproductions of these systems, and their published scores are not directly compared with ours because protocols differ.

No author names or institutional affiliations are asserted. The manuscript requires researcher and advisor review before submission.

3. Data and experimental design

We use UCI HAR [1]. The dataset provides pre-windowed 128-sample inertial sequences at 50 Hz, with six activity classes. We retain total accelerometer and gyroscope XYZ channels; derived body-acceleration channels are excluded. Official training subjects are subdivided into 17 training and four validation subjects using split seed 2026. Validation subjects are 26, 19, 25 and 1. The official nine test subjects are untouched during optimization. The resulting sample counts are 5,844 training, 1,508 validation and 2,947 test windows. All normalization statistics are computed exclusively from training windows.

A teaching mirror was used after slow access to the canonical UCI endpoint. Archive provenance and SHA-256 are supplied; dimensions and subject identities were verified, but byte identity to a completed origin download was not independently established. No raw data are redistributed with the code package.

Training-label fractions are 1%, 5%, 10% and 100%. Each class contributes ceil(fraction × class count), with a minimum of one window. A fixed permutation per class and seed produces nested subsets across fractions and identical subsets across methods. Existing ground-truth labels are used to construct these simulated stratified budgets. All validation labels remain available for model selection: fractions refer only to training labels, not total annotation costs. Overlapping windows within a subject are not treated as independent human participants.

Partition	Subjects	Windows
Training	17	5,844
Validation	4	1,508
Test	9	2,947

Nominal training labels	Actual labeled windows
1%	61
5%	295
10%	586
100%	5844

Archive SHA-256: 50dabbc800629611831a85b8b71c87040525ca4af6a152c6ba9360ecee6b92dc

4. Architecture and optimization

The encoder comprises Conv1D(6,32,k=7), GroupNorm(4,32), GELU, Conv1D(32,64,k=5), GroupNorm(8,64), and GELU. Same padding preserves temporal length. Global average pooling and a six-output linear layer perform classification. The classifier has 12,262 trainable parameters. A 1×1 convolution maps encoder outputs to six reconstruction channels during pretraining.

Given normalized signal x, a Boolean mask m marks missing positions, and the corrupted input is x multiplied by (1−m). The reconstruction objective is the mean squared error at masked positions only. Temporal masking removes one contiguous 32-sample span across all channels. Channel masking removes one randomly selected channel over the whole window. Combined masking takes their union. Their masked proportions are respectively 25%, 16.67% and 37.5%; this experiment does not isolate mask geometry from total coverage.

The supervised baseline starts from a randomly initialized encoder. Three SSL variants pretrain on all training windows with label targets excluded. Each uses 30 pretraining epochs; compatible pretrained encoders are reused across label budgets. All methods then train for 30 classification epochs with Adam, learning rate 0.001, batch size 128, without a learning-rate schedule. The classifier head initialization and shuffled labeled-example order are paired by seed. The encoder is fully fine-tuned. Checkpoints are selected by clean validation Macro-F1, with the first checkpoint retained in a tie. No test score changes hyperparameters or checkpoint selection. SSL has additional pretraining computation; the comparison is not compute matched.

Pretraining objective	Masked fraction	Geometry
Time	25%	32 contiguous time steps, all channels
Channel	16.67%	One of six channels, all time steps
Combined	37.5%	Union of time and channel masks

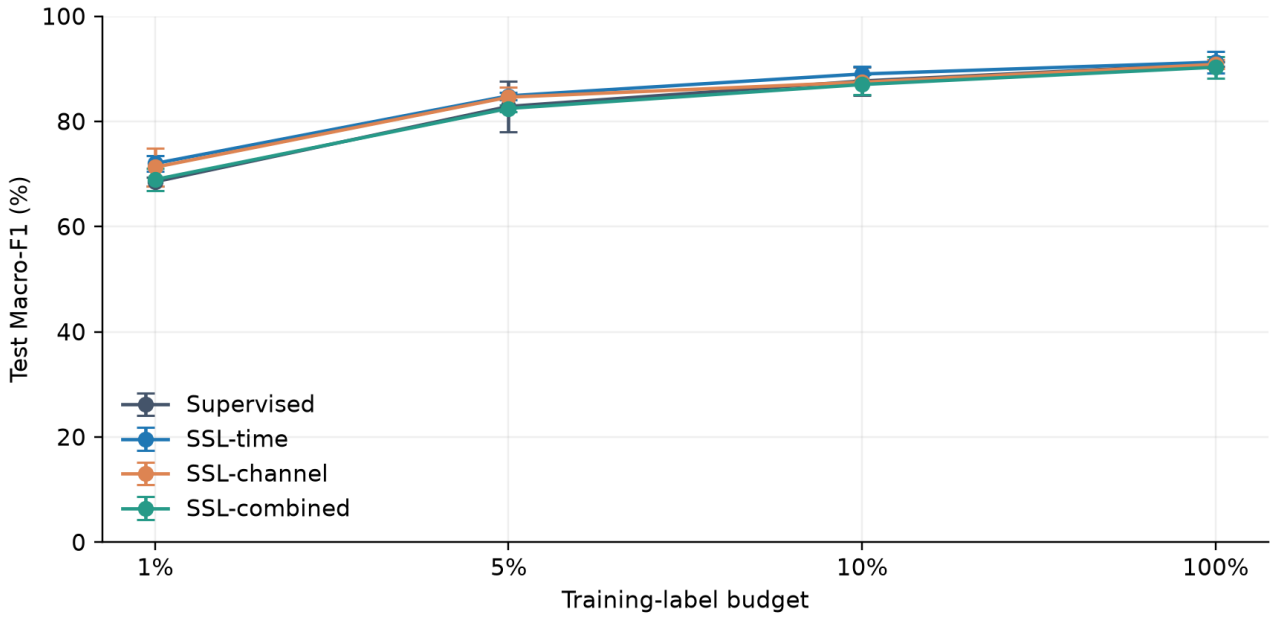
Three seeds (0,1,2), four label budgets and four methods yield 48 training configurations. Each selected checkpoint is evaluated on clean test windows, all accelerometer channels replaced by their training mean, and all gyroscope channels replaced by their training mean. These are simulated group outages, not observed hardware failures. Normalized zero replacement uses training statistics, with no test-time refitting.

The masked MSE objective is averaged over masked values only. Reconstruction quality is not itself an activity-recognition metric.

5. Clean-signal results

Macro-F1 (%), mean +/- sample standard deviation across seeds 0, 1 and 2. Every row uses the same labeled examples across methods.

Labels	Supervised	SSL-time	SSL-channel	SSL-combined
1%	68.52 +/- 0.84	72.00 +/- 1.47	71.30 +/- 3.58	68.95 +/- 2.06
5%	82.85 +/- 4.79	84.86 +/- 0.72	84.61 +/- 1.86	82.46 +/- 0.64
10%	87.68 +/- 2.68	89.01 +/- 1.47	87.46 +/- 0.29	87.01 +/- 1.87
100%	90.92 +/- 0.46	91.26 +/- 2.02	90.84 +/- 0.87	90.31 +/- 2.05



Paired effect of SSL relative to supervised training

Labels	SSL-time	SSL-channel	SSL-combined
1%	3.48 +/- 1.56	2.78 +/- 2.86	0.43 +/- 2.44
5%	2.01 +/- 5.02	1.76 +/- 3.24	-0.38 +/- 4.88
10%	1.33 +/- 1.25	-0.23 +/- 2.42	-0.67 +/- 3.21
100%	0.35 +/- 2.34	-0.07 +/- 1.31	-0.61 +/- 1.61

Paired differences are percentage points. Standard deviation is computed on the three per-seed differences, not by combining independent standard deviations.

6. Simulated sensor-outage results

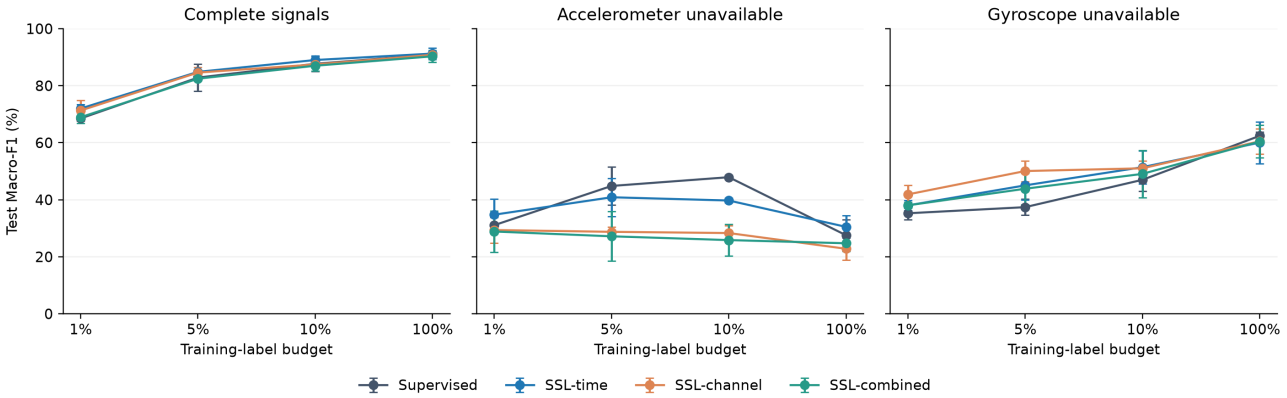
The selected clean-validation checkpoint is reused without further training. Each unavailable sensor group is replaced by its training mean.

Accelerometer unavailable

Labels	Supervised	SSL-time	SSL-channel	SSL-combined
1%	31.09 +/- 2.81	34.76 +/- 5.56	29.38 +/- 4.64	28.87 +/- 7.20
5%	44.84 +/- 6.75	40.88 +/- 6.69	28.75 +/- 1.77	27.18 +/- 8.71
10%	47.91 +/- 0.74	39.73 +/- 0.81	28.32 +/- 2.57	25.85 +/- 5.51
100%	27.54 +/- 5.48	30.49 +/- 3.94	22.80 +/- 4.01	24.74 +/- 2.86

Gyroscope unavailable

Labels	Supervised	SSL-time	SSL-channel	SSL-combined
1%	35.26 +/- 2.23	38.02 +/- 1.76	41.88 +/- 3.27	38.04 +/- 1.02
5%	37.40 +/- 2.80	45.05 +/- 5.08	50.07 +/- 3.63	43.86 +/- 5.47
10%	47.00 +/- 3.92	51.43 +/- 5.79	51.03 +/- 2.67	49.08 +/- 8.25
100%	62.42 +/- 1.32	60.05 +/- 7.32	60.51 +/- 4.46	60.44 +/- 5.76



Error bars show sample standard deviation across seeds. Channels are grouped into two physical sensor types. These stress tests do not represent every real-world missingness mechanism.

7. Exploratory test-subject uncertainty

We resample the nine test subjects with replacement 5,000 times, retaining all windows within each selected subject. Subject draws are paired across methods and seeds. The statistic is the mean of three seed-level differences in pooled-window Macro-F1. Percentile intervals are conditional on the fixed training split and fitted models.

Labels	Method	Difference (pp)	95% interval (pp)
1%	SSL-time	+3.48	[+0.77, +5.99]
1%	SSL-channel	+2.78	[+0.85, +4.63]
1%	SSL-combined	+0.43	[-2.27, +2.83]
5%	SSL-time	+2.01	[+0.02, +4.26]
5%	SSL-channel	+1.76	[-0.39, +3.72]
5%	SSL-combined	-0.38	[-2.90, +1.75]
10%	SSL-time	+1.33	[+0.43, +2.27]
10%	SSL-channel	-0.23	[-1.92, +1.44]
10%	SSL-combined	-0.67	[-2.13, +0.75]
100%	SSL-time	+0.35	[-0.42, +1.23]
100%	SSL-channel	-0.07	[-2.06, +1.78]
100%	SSL-combined	-0.61	[-1.88, +0.61]

This supplementary analysis was specified while the experiment matrix was running. It does not alter model fitting or selection. Only clean-condition differences are assessed. Nine subjects are few, no multiple-comparison correction is applied, and these intervals do not include training-split uncertainty. They must not be interpreted as a guarantee of population performance.

8. Interpretation and limitations

At 1% training labels, the highest observed mean clean Macro-F1 is 72.00% (SSL-time), +3.48 percentage points relative to supervised training. This is a descriptive test-set comparison, not a model-selection rule.

At 5% training labels, the highest observed mean clean Macro-F1 is 84.86% (SSL-time), +2.01 percentage points relative to supervised training. This is a descriptive test-set comparison, not a model-selection rule.

At 10% training labels, the highest observed mean clean Macro-F1 is 89.01% (SSL-time), +1.33 percentage points relative to supervised training. This is a descriptive test-set comparison, not a model-selection rule.

At 100% training labels, the highest observed mean clean Macro-F1 is 91.26% (SSL-time), +0.35 percentage points relative to supervised training. This is a descriptive test-set comparison, not a model-selection rule.

The study uses one dataset and one subject partition with three seeds. It does not reproduce contrastive or modern masked-model baselines, tune every method under an equal search budget, or match total computation. Full validation labels add annotation costs outside the advertised training-label budgets. Combined masking has a larger coverage fraction, and single-channel masking during pretraining differs from whole-group suppression during evaluation. Training windows are correlated and class-balanced subsampling assumes labels are already known. The protocol cannot establish generalization to new devices, placements, demographic groups, or clinical use. An empirical advantage would not establish architectural novelty or universal robustness; a negative result would not invalidate SSL in general.

Conclusion

This controlled experiment quantifies label-budget sensitivity and simulated sensor-outage behavior for a compact reconstruction-based SSL family. Any practical preference should consider validation evidence, seed variability, additional pretraining cost and the intended missing-sensor scenario. Broader claims require external baselines and independent datasets.

9. Reproducibility and submission status

Source, exact installed package versions, data checksum, split identities, labeled indices, learning curves, checkpoint weights, confusion matrices and per-class F1 are provided. A synthetic smoke test and unit tests check the pipeline separately from the research results. Generative AI assisted with code, analysis and manuscript preparation. The researcher and advisor must verify the work and comply with their institution and venue's disclosure rules before submission. No authorship, affiliation, ethics approval or acceptance is asserted.

Training completed on CPU with Python 3.14.6, PyTorch 2.14.1 and NumPy 2.5.4.

Training source commit: a96df30d8c308edc33b0113d6124420fa23e2d6e

References

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[4] Wang, J., Zhu, T., and Ning, H. (2023). An Improved Masking Strategy for Self-supervised Masked Reconstruction in Human Activity Recognition. arXiv:2312.04147. https://arxiv.org/abs/2312.04147

[5] Liu, X., Yuan, G., Zhang, Y., Liu, S., and Yan, Q. (2025). RobustHAR: Multi-scale Spatial-temporal Masked Self-supervised Pre-training for Robust Human Activity Recognition. IJCAI, 8563-8571. https://www.ijcai.org/proceedings/2025/0952.pdf

What remains before formal submission

Confirm the target venue and author details; agree on a defensible contribution with the advisor; add modern external baselines, additional datasets and subject splits; audit AI-use disclosure and source licensing. This package is a complete bounded experiment, not a guarantee of thesis completion or publication acceptance.

Appendix A. Complete clean-test runs

One row per configuration. Outage scores, per-class F1 and confusion matrices are included in the machine-readable artifacts.

Method	Labels	Seed	Macro-F1	Accuracy	Val F1
Supervised	1%	0	67.55	71.87	74.33
Supervised	1%	1	68.98	70.07	71.01
Supervised	1%	2	69.04	70.75	71.17
Supervised	5%	0	78.03	79.06	89.27
Supervised	5%	1	87.61	87.92	91.10
Supervised	5%	2	82.91	83.24	89.66
Supervised	10%	0	88.86	89.01	94.19
Supervised	10%	1	89.57	89.75	94.86
Supervised	10%	2	84.61	84.87	95.38
Supervised	100%	0	90.46	90.50	96.53
Supervised	100%	1	91.38	91.38	96.05
Supervised	100%	2	90.91	90.94	97.57
SSL-time	1%	0	71.77	73.16	77.03
SSL-time	1%	1	70.67	71.16	73.45
SSL-time	1%	2	73.57	74.31	81.97
SSL-time	5%	0	85.44	85.65	89.87
SSL-time	5%	1	85.08	85.34	92.50
SSL-time	5%	2	84.05	84.42	90.64
SSL-time	10%	0	89.90	89.96	95.91
SSL-time	10%	1	89.82	89.96	94.92
SSL-time	10%	2	87.32	87.48	93.08
SSL-time	100%	0	93.44	93.42	96.08
SSL-time	100%	1	90.91	90.87	95.30
SSL-time	100%	2	89.44	89.41	94.59
SSL-channel	1%	0	67.59	70.17	74.33
SSL-channel	1%	1	74.72	75.87	74.44
SSL-channel	1%	2	71.58	72.51	75.63
SSL-channel	5%	0	83.42	83.71	89.20
SSL-channel	5%	1	86.75	86.97	93.84
SSL-channel	5%	2	83.66	83.88	93.04
SSL-channel	10%	0	87.46	87.58	95.22
SSL-channel	10%	1	87.74	87.99	93.48
SSL-channel	10%	2	87.17	87.31	95.45
SSL-channel	100%	0	91.85	91.86	95.32
SSL-channel	100%	1	90.25	90.13	97.31
SSL-channel	100%	2	90.43	90.40	96.70
SSL-combined	1%	0	69.57	71.97	79.05
SSL-combined	1%	1	70.63	70.85	76.23
SSL-combined	1%	2	66.65	66.71	75.75
SSL-combined	5%	0	82.88	83.10	89.01
SSL-combined	5%	1	82.78	83.00	95.87
SSL-combined	5%	2	81.73	81.98	91.93
SSL-combined	10%	0	85.05	85.24	94.87
SSL-combined	10%	1	88.77	88.97	95.27
SSL-combined	10%	2	87.21	87.34	95.41
SSL-combined	100%	0	88.60	88.70	95.51
SSL-combined	100%	1	92.58	92.50	95.25
SSL-combined	100%	2	89.73	89.85	97.31